

Fairness in Credit Model:

The example of German Credit Data

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1. Introduction: Business Idea

Machine learning techniques have revolutionised many applications, significantly reducing the time and increasing the accuracy and efficacy of tasks traditionally performed by humans. The performance advantages and potential for immediate profits encourage businesses to adopt these techniques wherever possible, especially in heavily regulated areas where model output can profoundly impact people's lives, health, and well-being. However, integrating machine learning models into such sensitive and critical domains, emphasised the importance of addressing the problem of algorithmic bias and fairness. Negligence of these aspects poses severe legal and financial risks to all stakeholders involved and can be severely penalised.

One particular class of algorithmic solutions which are put into question are credit scoring models, employed by financial institutions to evaluate applicants' creditworthiness. These models are expected to be developed to classify the users regardless of their gender, race, sex or any other protected parameters, across all demographic groups. If these models are biased, they can unfairly disadvantage certain groups based on any of the protected attributes, leading to unequal access to credit and financial opportunities.

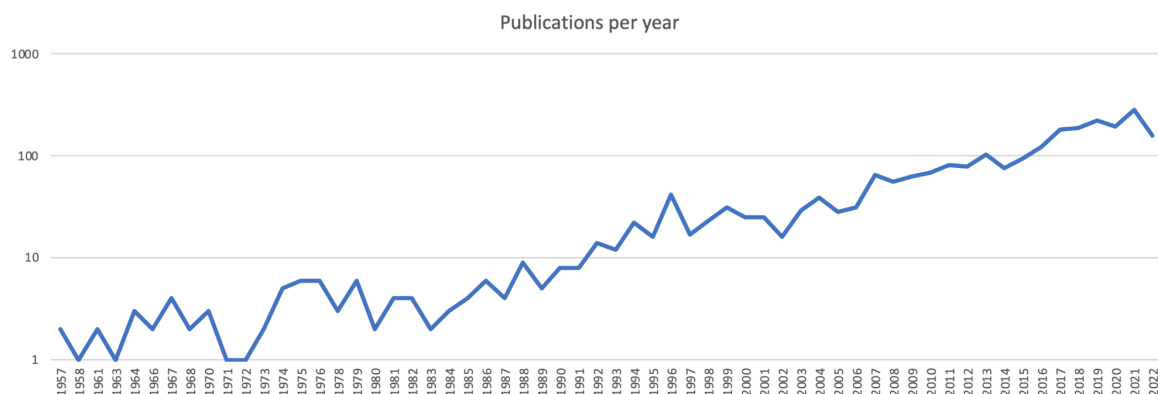


Figure 1: Publications on diversity and inclusion in management and business literature per year on a log(10) scale¹

¹

https://www.researchgate.net/publication/373938647_Prominent_Themes_and_Blind_Spots_in_Diversity_and_Inclusion_Literature_A_Bibliometric_Analysis/figures?lo=1

The fairness of credit models is crucial for businesses as it ensures compliance with anti-discrimination laws, helps avoid legal repercussions, and fosters trust in automated systems. However, this is just one of the reasons for its importance. From an economic and social perspective, fairness in credit models promotes inclusion and equity, providing equal opportunities for all individuals to access essential services. Additionally, fairness and equity can be pivotal in marketing, serving as both a “deal maker” and “deal breaker”, as the business world becomes increasingly concentrated on diversity and inclusion. (consult Figure 1 on page 2).

This report summarises an exemplary application of methods addressing the issue of gender bias in credit scoring models with the use of German Credit Data. The objective was to develop a model which not only minimises bias and discrimination but also upholds the best accuracy in terms of performance. A solution resting on our findings could be implemented in a real-life business setting.

2. Methodology: employing Data Science

The empirical case study involved utilising the German Credit data with the label describing client status (good/bad) to train the credit model and an optimal complementary bias mitigation algorithm. The bias pertained to the difference between the credit assessment of two groups: males and females.

Four distinct algorithms were used for handling algorithmic bias in three phases of model development: preprocessing, in-processing and post-processing. All of the methods were tested with the use of the same predictive model. After training, a comparison of the algorithms with respect to selected fairness and performance metrics was performed on a hold-out test set. All tested approaches were benchmarked against the baseline model. This model was optimised solely for the performance without the application of any fairness algorithms. The fairness of the original unprocessed labels was also assessed.

This comparative analysis provides a great insight into the effectiveness of different bias mitigation algorithms and their effect on predictive performance. The next few sections describe utilised data and algorithms in more detail.

2.1. Data selection and preparation

The German Credit dataset was downloaded in preprocessed format from [UC Irvine Machine Learning Repository](https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data)² and utilised for the development of the credit scoring model. The features consisted of categorical and integer variables describing the credit history and demographics of credit applicants with an initial feature count of 20. After one-hot encoding was applied to categorical features, the feature count expanded to fifty-eight. The encoded data was then divided into training and testing sets, where 70% of the observations were used to train the model and 30% for testing. Below we present the list of the 20 original features:

²

<https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>

Nominal Features:

- Status of existing checking account
- Credit history
- Purpose of the loan
- Savings account/bonds
- Present employment since
- **Personal status (4 values) recoded to male/female**
- Other debtors/guarantors
- Property (car, real estate)
- Other instalment plans (banks, stores)
- Housing (rent, own, free)
- Job
- Telephone
- Foreign worker

Numerical Features:

- Duration in month
- Credit amount
- Installment rate as % of disposable income
- Present residence since
- Age in years
- Number of existing credits at this bank
- Number of people being liable to provide maintenance for

2.2. Bias mitigation techniques

To ensure the fairness of the credit scoring model, the algorithms from the AIF360³ open source library were employed. This library contains techniques which were developed by the research community and IBM to help detect and mitigate the bias in machine learning. From this library, four algorithms were used. These are briefly discussed in the following sections.

2.2.1. Preprocessing algorithm: Reweighing

To ensure fairness before model training and development, the Reweighing preprocessing technique was used on the training data. This technique weights the

3

<https://aif360.readthedocs.io/en/latest/Getting%20Started.html>

examples in each combination of privileged classes and labels differently to ensure unbiased model training and classification. The primary metric optimised by reweighing is demographic parity or group fairness. Reweighing assists in creating a weighted dataset where the positive outcome rates are balanced across all groups, thus helping to promote demographic parity.

class

```
aif360.algorithms.preprocessing.Reweighing(unprivileged_groups, privileged_groups)4
```

2.2.2. In-processing algorithm: Grid-Search Reduction

Grid search is a technique of in-processing bias reduction. The algorithm assigns a cost factor with respect to all combinations of protected features and labels and solves an optimisation problem of finding a deterministic classifier with the lowest empirical error subject to fair classification constraints. The search is conducted among different values of factors describing classification cost for each combination of protected attributes. Here it was optimised with respect to disparate ratio (demographic parity).

class

```
aif360.algorithms.inprocessing.GridSearchReduction(estimator, constraints, prot_attr=None, constraint_weight=0.5, grid_size=10, grid_limit=2.0, grid=None, drop_prot_attr=True, loss='ZeroOne', min_val=None, max_val=None)5
```

2.2.3. In-processing algorithm: Adversarial Debiasing

Adversarial debiasing is another in-processing technique. Two models based on neural nets are used: an actual classifier and an adversary trained to determine protected attributes from predictions. The actual classifier is trained to maximise the accuracy of predictions and simultaneously reduce an adversary's capability to determine the protected attribute from the predictions. Procedure should lead to a

⁴

<https://aif360.readthedocs.io/en/latest/modules/generated/aif360.algorithms.preprocessing.Reweighing.html#aif360.algorithms.preprocessing.Reweighing>

⁵

<https://aif360.readthedocs.io/en/latest/modules/generated/aif360.algorithms.inprocessing.GridSearchReduction.html#aif360.algorithms.inprocessing.GridSearchReduction>

fair classifier as the group discrimination information is not carried by the classifier for the adversary to exploit.

class

```
aif360.algorithms.inprocessing.AdversarialDebiasing(unprivileged_groups,privileged_groups,scope_name,sess,seed=None,adversary_loss_weight=0.1,num_epochs=50,batch_size=128,classifier_num_hidden_units=200,debias=True)6
```

2.2.4. Post-processing algorithm: Equalized Odds Postprocessing

Equalized Odds Postprocessing is a technique that solves a linear program to find probabilities with which to change output labels to optimise equalised odds.

class

```
aif360.algorithms.postprocessing.EqOddsPostprocessing(unprivileged_groups,privileged_groups,seed=None)7
```

2.2.5. Fairness evaluation metrics

The fairness evaluation metrics are used to assess the fairness of the machine learning models. These metrics assist in ensuring that the models are not inadvertently biased and do not discriminate against any particular protected group based on any attributes including gender, race or age. Mentioned below are the metrics used for model evaluation in this study:

1. **Demographic parity:** Demographic parity requires that the likelihood of a positive outcome should be the same across different groups. This can be formulated as:

$$P(\hat{Y} = 1 | A = a) = P(\hat{Y} = 1 | A = b)$$

for any two given groups a and b , where the $\hat{Y}$ is the outcome which is predicted.

⁶

<https://aif360.readthedocs.io/en/latest/modules/generated/aif360.algorithms.inprocessing.AdversarialDebiasing.html>

⁷

<https://aif360.readthedocs.io/en/latest/modules/generated/aif360.algorithms.postprocessing.EqOddsPostprocessing.html>

2. **Equal opportunity difference:** Equal opportunity difference is a fairness metric used to measure the difference in true positive rates (TPR) between privileged and unprivileged groups. This metric ensures that a model is equally effective at correctly identifying positive instances for different groups. In other words, it focuses on whether the model gives an equal chance to all groups to achieve a positive outcome when they deserve it. Equal opportunity difference is expressed as:

$$TPR_{privileged} - TPR_{unprivileged}$$

3. **Average odds difference:** This metric is the arithmetic average difference in false positive rates and true positive rates between groups:

$$\frac{1}{2} [(FPR_{D=unprivileged} - FPR_{D=privileged}) + (TPR_{D=unprivileged} - TPR_{D=privileged})]$$

2.2.6. Performance evaluation metrics

Evaluating the performance of machine learning models is crucial to ensure they are effective and reliable. Here, we discuss two key performance metrics: Accuracy and False Positive Rate (FPR).

1. **Accuracy:** Accuracy is the ratio of correctly predicted instances to the total instances. It is a fundamental metric that provides a quick snapshot of how well the model is performing overall.

$$\frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

2. **FPR (False Positive Rate):** False Positive Rate (FPR) measures the proportion of actual negatives that are incorrectly classified as positives by the model. It is also known as the fall-out and is important from a creditworthiness modelling perspective as misclassifying bad clients is associated with higher cost than misclassifying good clients.

$$\frac{FP}{FP+TN}$$

2.3. Machine Learning model development

The empirical study was performed with the development of the CatBoost (Categorical Boosting) machine learning algorithm. It is a high-performance, open-source gradient boosting library developed by Yandex⁸, specifically designed to handle categorical data efficiently and making it particularly useful for tabular datasets with a mix of categorical and numerical features. After running the GridSearchCV to find the optimum hyperparameters, *auto_class_weights* = 'Balanced' was selected together with the default values of the remaining. Parameters ascribing weights to classes in case of imbalanced datasets usually ensure higher performance and increased sensitivity for positive examples which is recommended in credit scoring models.

3. Results

Below we demonstrate the values of fairness and performance metrics for all tested fairness algorithms on test data. The utilised model was Catboost with hyperparameters verified with a Grid-Search cross-validation.

Metric	Benchmark		Pre-Processing	In-Processing		Post-Processing
	Dataset	Performance-only model	Reweighting	Grid-Search Reduction	Adversarial Debiasing	Equalised Odds Postprocessing
Demographic parity	1.28	1.15	1.03	1.02	1.07	1.07
Equal opportunity difference	not applicable	-0.02	-0.01	-0.1	0.03	0.03
Average odds difference	not applicable	0.00	0.00	0.05	0.02	0.02

Table 1: Fairness evaluation metrics comparison

⁸ Xu, R., Chen, Y., Xiao, T., Wang, J., & Wang, X. (2021). Predicting the trend of stock index based on feature engineering and CatBoost model. *International Journal of Financial Engineering*, 8(02), 2150027.

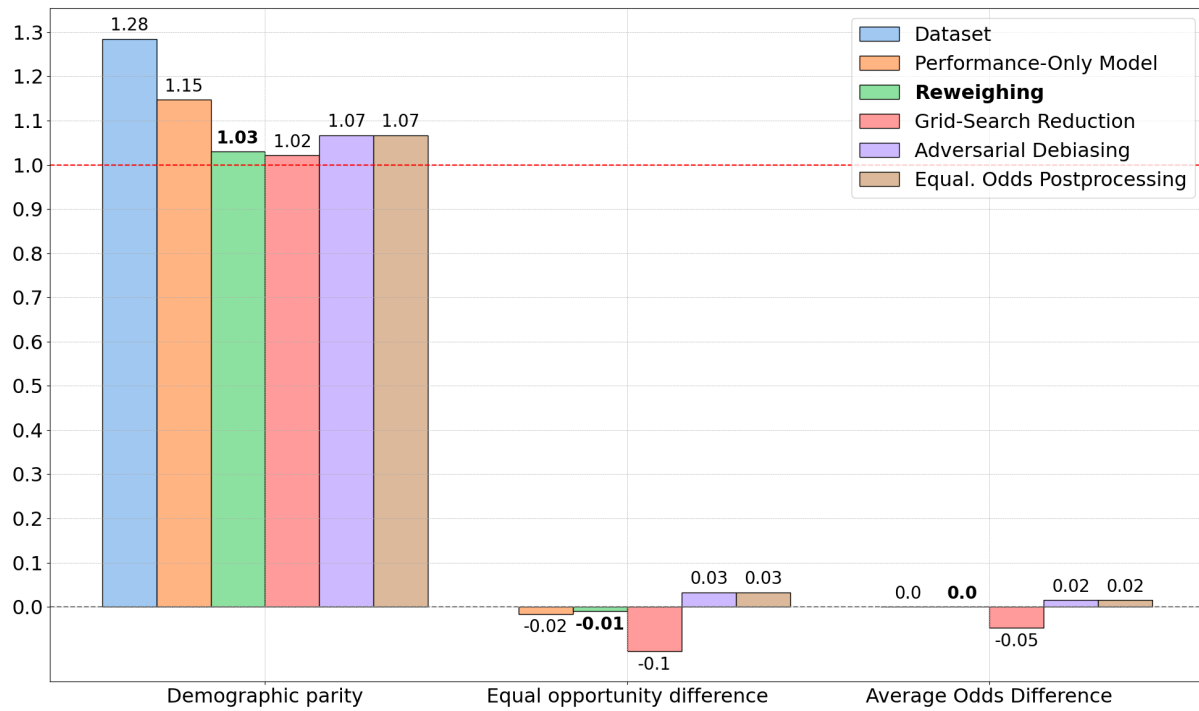


Figure 2: Fairness evaluation metrics comparison

Metric	Benchmark	Preprocessi ng	In processing		Postprocessing
	Performance- only model	Reweighing	Grid-Search Reduction	Adversarial Debiasing	Equalised Odds Postprocessing
Accuracy	0.77	0.77	0.76	0.64	0.64
FPR	0.17	0.17	0.18	0.22	0.45

Table 2: Performance evaluation metrics comparison

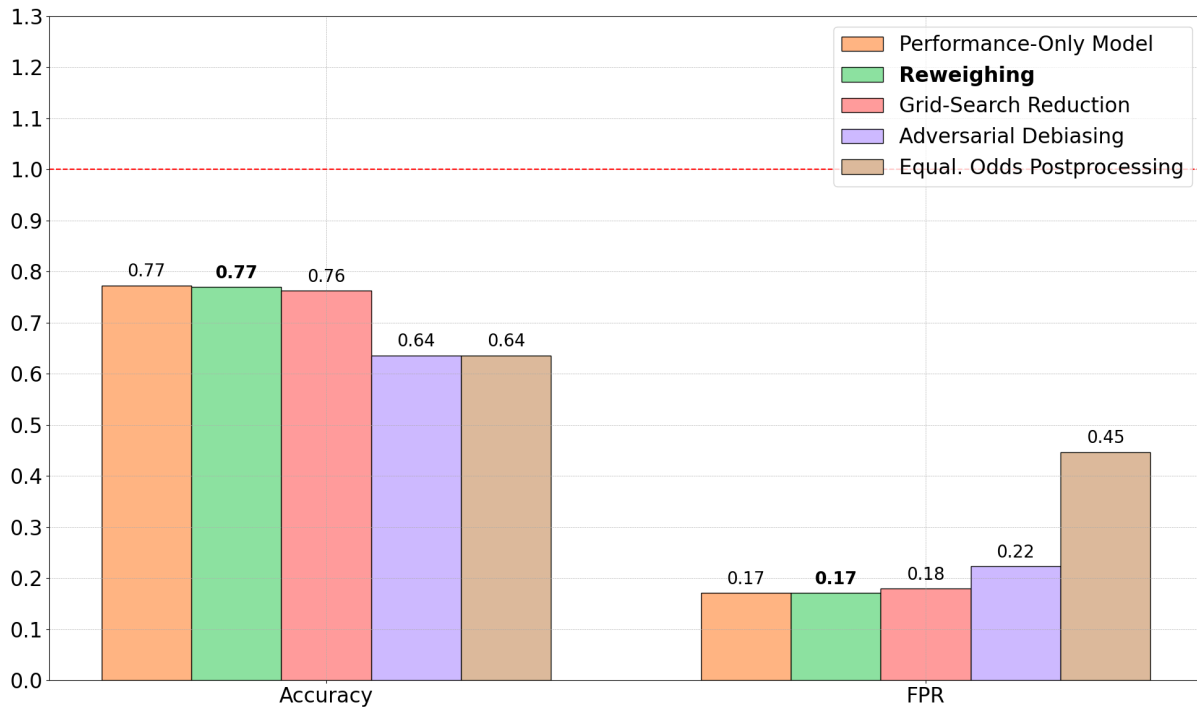


Figure 3: Performance evaluation metrics comparison

4. Discussion

From Table 1 and Table 2 and respective figures it can be concluded that the preprocessing bias reduction algorithm based on Reweighing provides the best balance maintaining the high accuracy of 0.77, which is similar to the performance-only model, and simultaneously improving the fairness metrics. It reduces the demographic parity to 1.03 compared to the performance-only model for which it was 1.15. It almost eliminates the equal opportunity difference (-0.01) while maintaining no average odds difference. Additionally, the FPR is achieved to be 0.17, thus making it the best-fitted technique, which not only significantly reduces the bias of the model but also upholds the accuracy of the algorithm without significant loss of predictive power.

While in-processing and post-processing methods also improve fairness metrics, they come with a considerable trade-off in terms of model accuracy and increased FPR. Therefore, the Reweighing method in the pre-processing stage stands out as the most effective approach for achieving a fair and high-performing credit scoring model.

5. Business implementation

After the descriptive analysis of the empirical study, it is very evident that the critical challenges of the financial industry in regards to ensuring the fairness of the model without hampering the predictive power can be addressed in some scenarios. Leveraging the capabilities of various bias mitigation techniques, we did an in-depth analysis for the best-suited process of bias mitigation in machine learning credit scoring algorithm. Our solution not only maximises the performance in terms of prediction but also significantly reduces the discriminative and biased characteristics of the algorithm, promoting ethical practices irrespective of gender.

Considering the works done by Köchling and Wehner (2020)⁹, Chang (2023)¹⁰, and articles related to Google Vision AI¹¹, algorithmic discrimination against women¹², there are a many cases where a corporation's reputation was not only damaged but also accompanied by a legal and financial risk due to the biased algorithms designed to inadvertently discriminate against protected features. Thus in a competitive world of finance, it is paramount to make credit scoring decisions that are both accurate and fair.

By proactively addressing biases, solutions similar to the one presented can help corporations ensure compliance with anti-discrimination laws and regulations, thereby mitigating legal and financial risks. Such a solution can also serve as valuable marketing resource by enhancing the bank's reputation. In the long run, unbiased, data-driven decision making is one of the best means of securing market trust, as well as a commitment to ethical practices.

⁹ <https://link.springer.com/article/10.1007/s40685-020-00134-w#Sec1>

¹⁰

https://www.researchgate.net/publication/373896468_Gender_Bias_in_Hiring_An_Analysis_of_the_Impact_of_Amazon's_Recruiting_Algorithm

¹¹ <https://algorithmwatch.org/en/google-vision-racism/>

¹² <https://theconversation.com/googles-algorithms-discriminate-against-women-and-people-of-colour-112516>

6. Scripts

Import of necessary packages

```
# Load all necessary packages
import sys
sys.path.insert(1, "../")

import numpy as np
np.random.seed(0)

import pandas as pd
pd.set_option('display.max_columns', None)

from aif360.datasets import GermanDataset
from aif360.metrics import BinaryLabelDatasetMetric, ClassificationMetric
from aif360.algorithms.inprocessing import GridSearchReduction
from aif360.algorithms.inprocessing import AdversarialDebiasing
# import tensorflow as tf
from IPython.display import Markdown, display

from sklearn.metrics import confusion_matrix
```

Data downloading, indicating privileged/unprivileged classes:

```
def map_label(df):
    # Define mapping dictionary
    mapping = {2.0: 1.0, 1.0: 0}

    # Map the 'credit' column
    df['credit'] = df['credit'].map(mapping)

    # Create 'sex' column based on 'personal_status'
    df['sex'] = df['personal_status'].apply(lambda x: 'female' if 'A92' in x else 'male')

    return df

dataset_orig = GermanDataset(
    protected_attribute_names=['sex'],
    privileged_classes=['male'], # male is considered privileged
    custom_preprocessing=map_label
```

```
)

dataset_orig_train, dataset_orig_test = dataset_orig.split([0.7], seed=12, shuffle=True)

privileged_groups = [{'sex': 1}]
unprivileged_groups = [{'sex': 0}]
```

Grid search for Catboost

```
from catboost import CatBoostClassifier
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
from sklearn.model_selection import GridSearchCV

cgb = CatBoostClassifier(
    eval_metric='AUC', # Use AUC for overfitting detector
    random_seed=42,
    verbose=0)

parameters = {
    'n_estimators': [50, 150],
    'max_depth': [5, 10],
    'learning_rate': [0.01, 0.5],
    'border_count': [254],
    'random_strength': [0.1, 0.5, 1],
    'bagging_temperature': [0.5, 1, 2],
    'l2_leaf_reg': [1, 3, 5]
}

grid_search = GridSearchCV(cgb, parameters, scoring='accuracy', cv=3, n_jobs=-1,
    verbose=3)
grid_search.fit(dataset_orig_train.features, dataset_orig_train.labels.ravel()) #
Assuming you have X_train and y_train defined

best_params = grid_search.best_params_
best_model = grid_search.best_estimator
```

Application of preprocessing algorithm: Reweighing

```
RW = Reweighing(unprivileged_groups=unprivileged_groups,
    privileged_groups=privileged_groups)
```

```

# compute the weights for reweighing the dataset
RW.fit(dataset_orig_train)

# transform the dataset to a new dataset based on the estimated transformation
dataset_rw_train = RW.transform(dataset_orig_train)
dataset_rw_test = RW.transform(dataset_orig_test)

print("Metrics for Original Training Dataset:")
print("\tDifference in mean outcomes between unprivileged and privileged groups = %f" %
metric_orig_train.mean_difference())
print("\tdisparate_impact = %f" % metric_orig_train.disparate_impact())

metric_rw_train = BinaryLabelDatasetMetric(dataset_rw_train,
                                           unprivileged_groups=unprivileged_groups,
                                           privileged_groups=privileged_groups)

print("Metrics for Reweighted Training Dataset:")
print("\tDifference in mean outcomes between unprivileged and privileged groups = %f" %
metric_rw_train.mean_difference())
print("\tdisparate_impact = %f" % metric_rw_train.disparate_impact())

```

Model development after Reweighting:

```

X_train_rw = dataset_rw_train.features
y_train_rw = dataset_rw_train.labels.ravel()
w_train_rw = dataset_rw_train.instance_weights.ravel()

X_test_rw = dataset_rw_test.features
y_test_rw = dataset_rw_test.labels.ravel()
w_test_rw = dataset_rw_test.instance_weights.ravel()

# Initialize the CatBoostClassifier with desired parameters
cgb_rw = CatBoostClassifier(auto_class_weights='Balanced', verbose=0)

# Train the CatBoostClassifier model on the preprocessed training data
cgb_rw.fit(X_train_rw, y_train_rw)

y_train_pred_rw = cgb_rw.predict(X_train_rw)

# assign positive class index
pos_ind_rw = np.where(cgb_rw.classes_ == dataset_rw_train.favorable_label)[0][0]

# add predicted labels to predictions dataset
dataset_rw_train_pred = dataset_rw_train.copy()
dataset_rw_train_pred.labels = y_train_pred_rw

dataset_rw_test_pred = dataset_rw_test.copy()

```



```

X_test = scale_rw.transform(dataset_rw_test_pred.features)
y_test_pred = cgb_rw.predict(X_test)
pos_ind_test = np.where(cgb_rw.classes_ == dataset_rw_test.favorable_label)[0][0]
dataset_rw_test_pred.labels = cgb_rw.predict(X_test)

```

Application of in-processing algorithm: Grid-Search Reduction

```

cgb = CatBoostClassifier(auto_class_weights='Balanced', verbose=0, **best_params)
grid_reduce = GridSearchReduction(prot_attr='sex', estimator=cgb, constraints =
'DemographicParity')
grid_reduce.fit(dataset_orig_train)
dataset_redu_train = grid_reduce.predict(dataset_orig_train)
dataset_redu_test = grid_reduce.predict(dataset_orig_test)

```

Application of in-processing algorithm: Adversarial Debiasing

```

import tensorflow.compat.v1 as tf
tf.reset_default_graph()
sess = tf.Session()
tf.disable_eager_execution()

ad = AdversarialDebiasing(sess=sess,
                          unprivileged_groups=unprivileged_groups,
                          privileged_groups=privileged_groups,
                          scope_name="debiasing_classifier")

ad.fit(dataset_orig_train)
dataset_ad_train = ad.predict(dataset_orig_train)
dataset_ad_test = ad.predict(dataset_orig_test)
sess.close()

```

Application of post-processing algorithm:

```

dataset_orig_train, dataset_orig_test = dataset_orig.split([0.7], seed=12,
shuffle=True)
df_cat = dataset_orig_train.convert_to_dataframe()[0].drop(['credit'], axis=1)
credit_test = dataset_orig_test.convert_to_dataframe()[0].drop(['credit'],
axis=1)
credit_test_og = dataset_orig_test.convert_to_dataframe()[0]['credit']

df_cat['credit'] = cgb.predict(dataset_orig_train.features)
credit_test['credit'] = cgb.predict(dataset_orig_test.features)

```

```

dataset_catboost = BinaryLabelDataset(df = df_cat,
                                       label_names=['credit'],
                                       favorable_label=1,
                                       protected_attribute_names=['sex'],
                                       privileged_protected_attributes
= [['male']])
credit_test = BinaryLabelDataset(df = credit_test,
                                  label_names=['credit'],
                                  favorable_label=1,
                                  protected_attribute_names=['sex'],
                                  privileged_protected_attributes
= [['male']])
cost_constraint = 'weighted'

cpp = CalibratedEqOddsPostprocessing(privileged_groups = privileged_groups,
                                     unprivileged_groups = unprivileged_groups,
                                     cost_constraint=cost_constraint,
                                     seed=randseed)
cpp = cpp.fit(dataset_orig_train, dataset_catboost)
dataset_transf_train_pred = cpp.predict(dataset_orig_train)
dataset_transf_test_pred = cpp.predict(credit_test)

```

Calculation of metrics (on the example of Grid-Search Reduction)

```

# Metrics for the dataset from model with Grid-Search Reduction
display(Markdown("#### Model - best CatBoost + GRD - dataset metrics"))
metric_dataset_redu_train = BinaryLabelDatasetMetric(dataset_catboost_train,
                                                       unprivileged_groups=unprivileged_groups,
                                                       privileged_groups=privileged_groups)

print("Train set: Difference in mean outcomes between unprivileged and privileged groups
= %f" % metric_dataset_redu_train.mean_difference())

metric_dataset_redu_test = BinaryLabelDatasetMetric(dataset_catboost_test,
                                                       unprivileged_groups=unprivileged_groups,
                                                       privileged_groups=privileged_groups)

print("Test set: Difference in mean outcomes between unprivileged and privileged groups
= %f" % metric_dataset_redu_test.mean_difference())

display(Markdown("#### Model - best CatBoost- classification metrics"))
classified_metric_reduction_test = ClassificationMetric(dataset_orig_test,

```

```

dataset_catboost_test,
unprivileged_groups=unprivileged_groups,
privileged_groups=privileged_groups)

print("Test set: Classification accuracy = %f" %
classified_metric_reduction_test.accuracy())
TPR = classified_metric_reduction_test.true_positive_rate()
TNR = classified_metric_reduction_test.true_negative_rate()
FPR = classified_metric_reduction_test.false_positive_rate()
FNR = classified_metric_reduction_test.false_negative_rate()
bal_acc_reduction_test = 0.5*(TPR+TNR)
print("Test set: Balanced classification accuracy = %f" % bal_acc_reduction_test)
print("Test set: Balanced TPR, FPR = %f" % TPR, FPR)
print("Test set: Balanced TNR, FNR = %f" % TNR, FNR)
print("Test set: Disparate impact = %f" %
classified_metric_reduction_test.disparate_impact())
print("Test set: Equal opportunity difference = %f" %
classified_metric_reduction_test.equal_opportunity_difference())
print("Test set: Average odds difference = %f" %
classified_metric_reduction_test.average_odds_difference())
print("Test set: Theil_index = %f" % classified_metric_reduction_test.theil_index())

```

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